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Matter RGB Light with the Arduino Nano Matter

Learn how to build a Matter RGB light using the Arduino Nano Matter

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Overview

This tutorial will teach you how to use the Arduino Nano Matter to create a Matter RGB Lightbulb to light up any room with colors.



RGB light overview

Thanks to the seamless compatibility of the Nano Matter with almost any Matter network we can easily integrate our RGB light with Amazon Alexa, Google Assistant, Apple HomeKit, Home Assistant and even any professional custom solution.

We have prepared a short demo in video format in case you are a visual learner.

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Build a Matter-Enabled R...



Hardware and Software Requirements

Hardware Requirements

[Arduino Nano Matter](#) (x1)

Grove - 8x8 RGB LED Matrix (x1)

Breadboard (x1)

Jumper wires

Eero 6+ WiFi Extender (Thread
Border Router) (x1)

[USB-C® cable](#) (x1)

Software Requirements

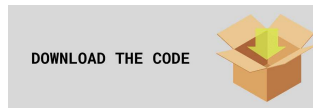
[Arduino IDE 2.0+](#) or [Arduino
Cloud Editor](#)

[Amazon Alexa](#)

[Seeed_RGB_LED_Matrix](#) library

You can install it as .ZIP using the Arduino IDE.

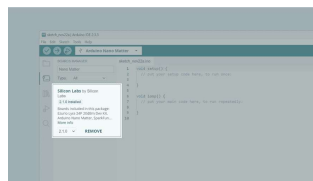
Download the Project Code



Download the complete project code [here](#).

Board Core and Libraries

The **Silicon Labs** core contains the libraries and examples you need to work with the board's components, such as its Matter, Bluetooth® Low Energy, and I/Os. To install the Nano Matter core, navigate to **Tools > Board > Boards Manager** or click the Boards Manager icon in the left tab of the IDE. In the Boards Manager tab, search for `Nano Matter` and install the latest `Silicon Labs` core version.

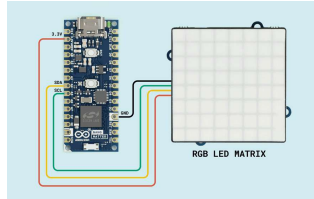


Installing the Silicon Labs core in the Arduino IDE

Project Setup

Schematic Diagram

Use the following connection diagram for the project:

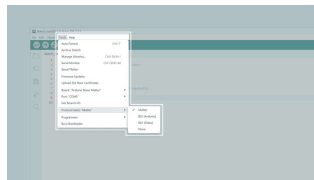


Project wiring diagram

The RGB LED matrix is powered by the Nano Matter `3.3V` output pin, and it is controlled using an I2C connection.

Programming

In the Arduino IDE upper menu, navigate to **Tools > Protocol stack** and select **Matter**.



Matter Protocol stack
selected

Copy and paste the following sketch

```

1  #include <Matter.h>
2  #include <MatterLightbulb.h>
3  #include "grove_two_rgb_led_matrix.h"
4
5  #define LED_R LED_BUILTIN
6  #define LED_G LED_BUILTIN
7  #define LED_B LED_BUILTIN
8
9  MatterColorLightbulb matter_color_bulb;
10
11  GroveTwoRGBLedMatrixClass matrix(LED_R, LED_G, LED_B);
12
13  void update_led_color() {
14  }
15  void led_off();
16  void handle_button_press();
17  volatile bool button_pressed = false;
18
19  void setup() {
20    Wire.begin();
21    Serial.begin(115200);
22    Matter.begin();
23    matter_color_bulb.begin(LED_R, LED_G, LED_B);
24    matter_color_bulb.brightness(100);
25
26    // Set up the onboard button
27    pinMode(BTN_BUILTIN, INPUT_PULLUP);
28    attachInterrupt(BTN_BUILTIN, handle_button_press, FALLING);
29  }
30
31  void loop() {
32    // Check if button is pressed
33    if (digitalRead(BTN_BUILTIN) == LOW) {
34      button_pressed = true;
35    }
36
37    // Update LED color
38    update_led_color();
39
40    // Turn off LED
41    led_off();
42  }

```

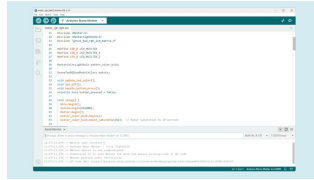
The structure of this example code is very simple, the main functions are explained below:

In the `setup()` function we initialize the Matter connectivity and the RGB Matrix I2C communication.

In the `loop()` function we listen to any request of controlling the lightbulb state or color and update the matrix respectively.

There are some other functions to handle button press or RGB matrix color setup.

Once you uploaded the example code to the Nano Matter, open the Serial Monitor and reset the board.



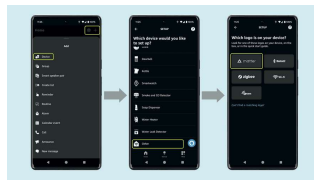
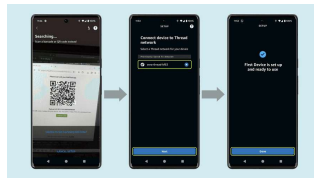
QR Code URL

After the reset you will find on the serial port the URL that generates the QR for the Matter device commissioning.

Adding the Device (Commissioning)

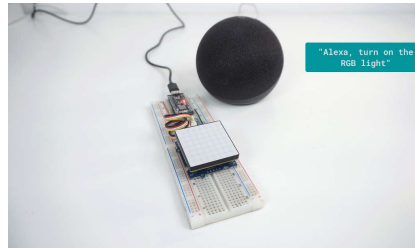
Copy and paste the QR code URL on your favorite web browser and a unique QR code will be generated for your board.

Go to your **Amazon Alexa** app, navigate to **Devices > Add Device > Other**, select the **Matter** option and scan the QR code.

Adding the device to
Google Home appYour device is successfully
added

Final Results

Finally, you will be able to control the RGB light from your smartphone, hub, or asking your personal assistant.



Conclusion

In this tutorial we have learned how to create a Matter enabled RGB light that can be controlled from our smartphone and personal assistant. The Nano Matter allows us to seamlessly integrate our own light a a commercial product with our current smart home ecosystem.

Next Steps

You can take this solution even further by adding fading animations or even using the matrix to display text or graphics.

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